

A final example of an adaptive voltage-follower amplifier is shown in Fig. 27.21. This amplifier can only source current to a load. If v_{in} and v_{out} are equal, the current that flows in M1 and M2 is $I_{SS} + I_{D6}$. If v_{in} is increased, the current in M1 and M3 increases. This causes the currents in M4–M6 to increase, effectively increasing the tail current of the diff-pair. The result is a large current available to drive the load. Note that M7 can be sized larger than the other MOSFETs to increase maximum output current.

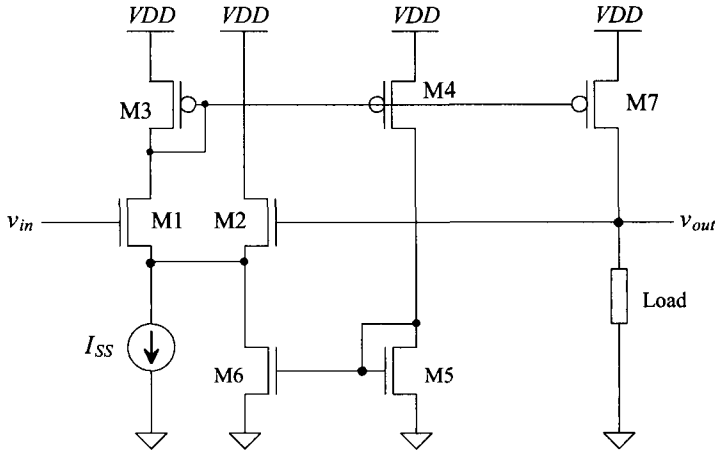


Figure 27.21 Adaptive voltage follower.

27.3 Analog Multipliers

Analog multipliers find extensive use in communication systems. Figure 27.22 shows the voltage characteristics of a four-quadrant multiplier. This multiplier is termed a four-quadrant multiplier because both inputs can be either positive or negative around a common-mode voltage, V_{CM} . The ideal output of the multiplier is related to the inputs by

$$v_{out} = K_m \cdot v_x v_y \quad (27.11)$$

where K_m is the multiplier gain with units of V^{-1} . In reality, imperfections exist in the multiplier gain, resulting in offsets and nonlinearities. The output of the multiplier can be written as

$$v_{out} = K_m(v_x + V_{OSx})(v_y + V_{OSy}) + V_{OSout} + v_x^n + v_y^m \quad (27.12)$$

where V_{OSx} , V_{OSy} , and V_{OSout} are the offset voltages associated with the x-, y-inputs, and the output, respectively. The terms v_x^n and v_y^m represent nonlinearities in the multiplier. Normally, these nonlinearities are specified in terms of the total harmonic-distortion or by specifying the maximum deviation in percentages between a straight line and the actual characteristic curves shown in Fig. 27.22 over some range of input voltages. Although many different techniques exist for implementing analog multipliers in CMOS, we concentrate on a technique useful in high- and low-frequency multiplication.

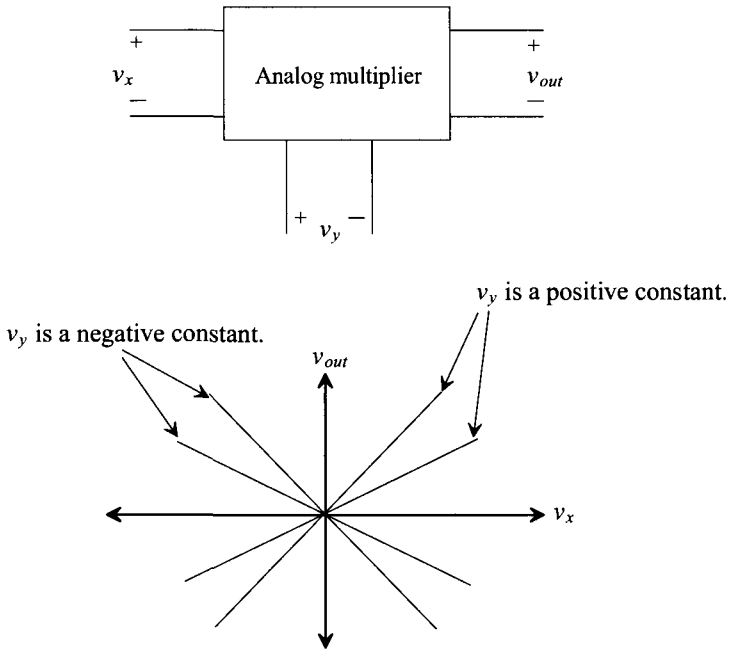


Figure 27.22 Operation of a four-quadrant analog multiplier.

27.3.1 The Multiplying Quad

A CMOS multiplier employing a multiplying quad (M1–M4) is shown in Fig. 27.23. The multiplying quad operates in the triode region, and thus MOSFETs M1–M4 can be thought of as resistors. For the moment we will not consider the biasing of the quad. The negative output voltage of the multiplier is given by

$$v_{om} = -R \cdot (i_{D1} + i_{D2}) \quad (27.13)$$

while the positive output voltage is

$$v_{op} = -R \cdot (i_{D3} + i_{D4}) \quad (27.14)$$

The output voltage of the multiplier is

$$v_{out} = v_{op} - v_{om} = R \cdot (i_{D1} + i_{D2} - i_{D3} - i_{D4}) \quad (27.15)$$

A simplified schematic of the multiplying quad with biasing is shown in Fig. 27.24. The op-amp inputs are at an AC virtual ground and at a DC voltage of V_{CM} (the op-amp output common-mode voltage). In order to minimize the DC input current on the x-axis inputs, the common-mode DC voltage on this input is set to V_{CM} . The DC biasing voltage on the y-input is set to a value large enough to keep the quad in triode. The input signals have been broken into two parts (e.g., $v_x/2$ and $-v_x/2$) to maintain generality. In practice, the minus inputs can be connected directly to the bias voltages at the cost of

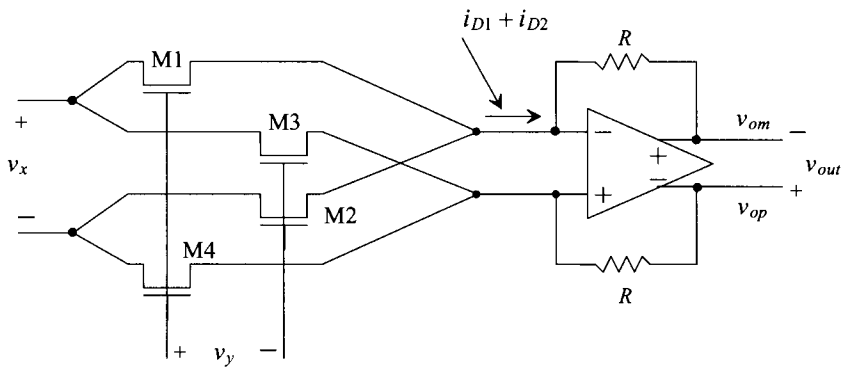


Figure 27.23 CMOS analog multiplier.

large-signal linearity. (The input is not truly differential in this situation.) However, as discussed earlier, a fully differential system has much better coupled noise immunity.

Using Eq. (9.12) and noticing that the DC gate-source voltage of all MOSFETs is the same, the drain currents can be written as

$$i_{D1} = \beta_1 \left[\left(V_{GS} + \frac{v_y}{2} - V_{THN1} \right) \left(\frac{v_x}{2} \right) - \frac{1}{2} \left(\frac{v_x}{2} \right)^2 \right] \quad (27.16)$$

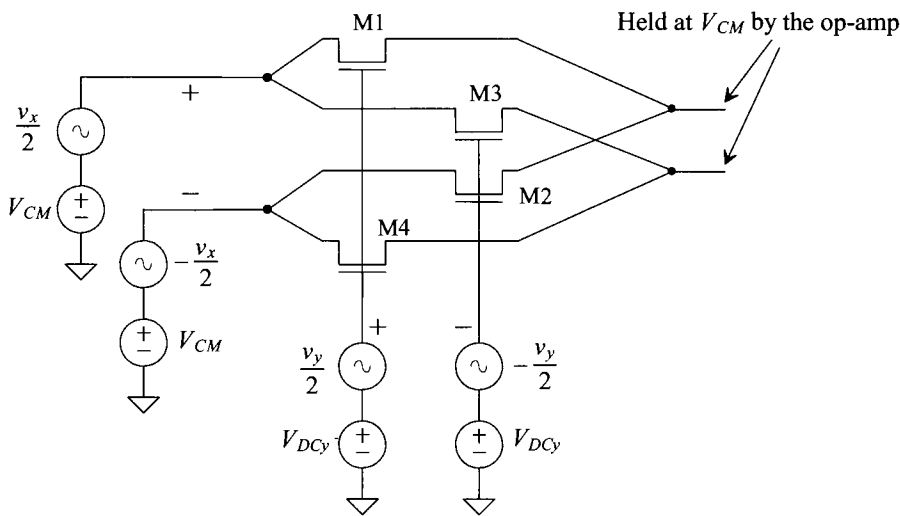


Figure 27.24 Biasing of the multiplying quad.

$$i_{D2} = \beta_2 \left[\left(V_{GS} - \frac{v_y}{2} - V_{THN2} \right) \left(-\frac{v_x}{2} \right) - \frac{1}{2} \left(-\frac{v_x}{2} \right)^2 \right] \quad (27.17)$$

$$i_{D3} = \beta_3 \left[\left(V_{GS} - \frac{v_y}{2} - V_{THN3} \right) \left(\frac{v_x}{2} \right) - \frac{1}{2} \left(\frac{v_x}{2} \right)^2 \right] \quad (27.18)$$

$$i_{D4} = \beta_4 \left[\left(V_{GS} + \frac{v_y}{2} - V_{THN4} \right) \left(-\frac{v_x}{2} \right) - \frac{1}{2} \left(-\frac{v_x}{2} \right)^2 \right] \quad (27.19)$$

We can design so that $\beta = \beta_1 = \beta_2 = \beta_3 = \beta_4$. We can use Eq. (27.15) together with Eqs. (27.16)–(27.19) to rewrite the output voltage of the multiplier as

$$v_{out} = R\beta \cdot \left(\frac{v_x}{2} \right) \left[\frac{v_y}{2} - V_{THN1} + \frac{v_y}{2} + V_{THN2} + \frac{v_y}{2} + V_{THN3} + \frac{v_y}{2} - V_{THN4} \right] \quad (27.20)$$

We can see that if $V_{THN1} = (V_{THN2} \text{ or } V_{THN3})$ and $V_{THN4} = (V_{THN3} \text{ or } V_{THN2})$, this equation can be rewritten as

$$v_{out} = R\beta \cdot v_x v_y \quad (27.21)$$

The source of a MOSFET (the terminal we label “source” depends on which way current flows in the MOSFET) in the multiplying quad is connected either to the op-amp or to the x inputs. When the sources of the MOSFETs are connected to the op-amp, all of the MOSFETs in the multiplying quad have the same threshold voltage. (Since the source of each MOSFET is tied to the same potential, the body effect changes each MOSFET’s threshold voltage by the same amount.) If the positive x-input is sinking a current, then the sources of M1 and M3 are the “+” x-input and thus $V_{THN1} = V_{THN3}$. In any case, the threshold voltages of the MOSFETs cancel and Eq. (27.21) holds. Comparing Eqs. (27.21) and (27.11) results in defining the gain of this multiplier as

$$K_m = R \cdot \beta \quad (27.22)$$

Simulating the Operation of the Multiplier

Simulating the performance and understanding the analog multiplier operation is an important step in the design process. The design of a multiplier consists of designing the op-amp, selecting the sizes of the multiplying quad, and designing the biasing network. Because we covered the design of differential input/output op-amps in the last chapter, it is not covered here. In order to simulate the performance of a multiplier in SPICE without including the limitations of the op-amp, the simple model shown in Fig. 27.25 is used. The sum of the multiplying factors associated with the voltage-controlled voltage sources, E1 and E2, is the open-loop gain of the op-amp. A typical SPICE statement for these VCVS (voltage-controlled voltage source) where the op-amp open loop gain is 20,000 is

```
E1      Voplus 8 4 3   1E4
E2      8 Vominus 4 3 1E4
```

where the nodes correspond to those labeled in Fig. 27.25.

The next problem we encounter in simulating the operation of the multiplier is implementing the differential voltages (e.g., $\pm v_x/2$), in addition to the DC biasing voltages. The setup shown in Fig. 27.26 is used to implement the biasing and the differential voltage sources. The op-amp common-mode output voltage, V_{CM} , and the x

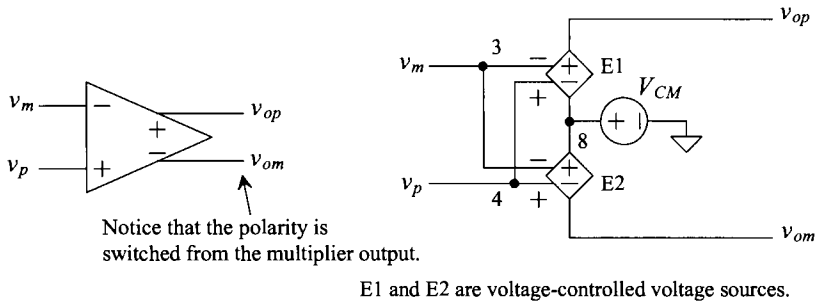


Figure 27.25 SPICE modeling a differential input/output op-amp with common-mode voltage.

input voltage are set to 1.5 V. The lower this voltage, the easier it is to bias the multiplying quad into the triode region. On the other hand, a reduction in the value of V_{CM} limits the op-amp output voltage swing and thus the multiplier output range. The size of the multiplying quad was set to 10/2. The larger the W/L ratio of the MOSFETs used in this quad, the easier it is to keep the quad in the triode region. On the other hand, using a large W/L increases the required input current. The channel length can be increased to ensure the device operates as a long-channel device and follows Eq. (9.12). Since the quad is part of the feedback around the op-amp, long-channel devices do not affect the

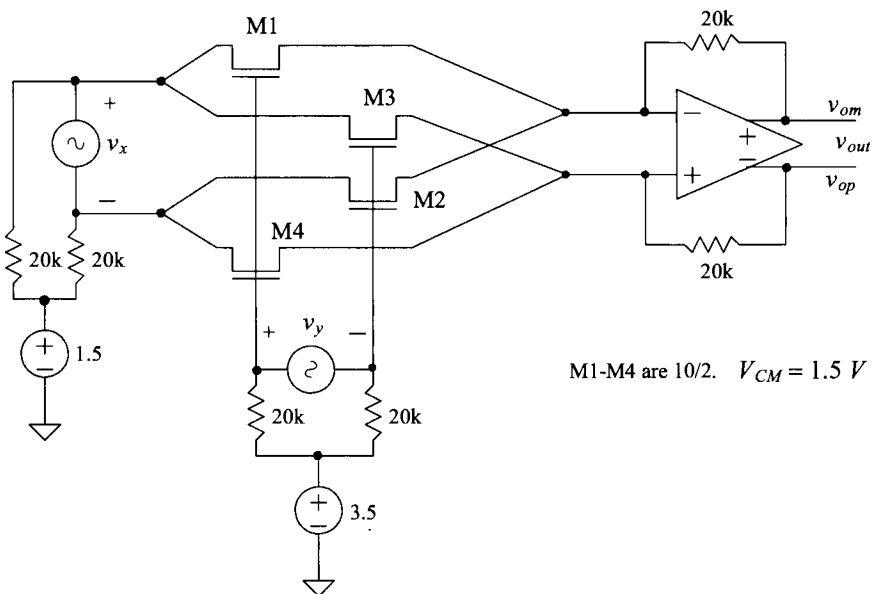


Figure 27.26 SPICE simulation schematic.

speed. The DC voltage at the y-inputs was set to 3.5 V as a compromise between keeping the multiplying quad in triode and the y-input voltage range. The gain of the multiplier in Fig. 27.26 is, from Eq. (27.22) and Table 6.2,

$$K_m = 20k \cdot 120 \frac{\mu\text{A}}{\text{V}^2} \cdot \frac{10}{2} = 12 \text{ V}^{-1}$$

A DC sweep showing the operation of the multiplier is shown in Fig. 27.27. The x-input, v_x , was swept from -1 to $+1$, while at the same time the y-input was stepped from -1 to 1 V in 0.5 V increments. Keeping in mind that the output of the multiplier is $v_{op} - v_{om}$, we can understand the data presented in Fig. 27.27 by considering points A and B. At point A, the y-input is 1 V while the x-input is 0.25 V. The output voltage of the multiplier is the product of the multiplier gain and these two voltages (i.e., $12 \cdot 1 \cdot 0.25 = 3$ V). The output voltage at point B is $12 \cdot 0.5 \cdot (-0.3) = -1.8$ V. Note that this figure was generated using an almost ideal op-amp. The characteristics do not show the limitations of the op-amp. In particular, the limited output swing.

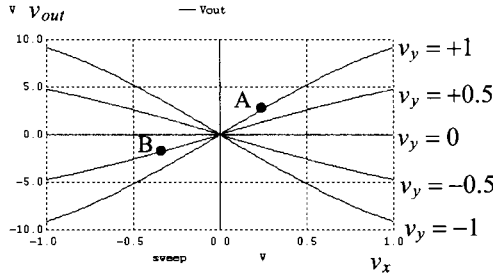


Figure 27.27 DC characteristics of the multiplier of Fig. 26.26.

27.3.2 Multiplier Design Using Squaring Circuits

An analog multiplier can be designed based on the difference between the sum of two voltages squared and the difference of two voltages squared, or

$$V_o = (V_1 + V_2)^2 - (V_1 - V_2)^2 = 4V_1V_2 \quad (27.23)$$

The basic sum-squaring and difference-squaring circuits are shown in Fig. 27.28. MOSFETs M1 and M4 are source-followers, while MOSFETs M2 and M4 are called squaring MOSFETs. This circuit is designed so that $\beta_1 = \beta_4 = \beta_{14}$, $\beta_2 = \beta_3 = \beta_{23}$, and $\beta_{14} \gg \beta_{23}$. This makes almost all of the DC bias currents, I_{S12} and I_{S34} , flow in the MOSFETs M1 and M4, respectively. The squaring current, I_{SQ} , assuming that zero current flows through the resistor when both inputs are zero volts (or whatever the common-mode voltage when a single supply is used), is given by

$$I_{SQ(a)} = \frac{\beta_{23}}{4}(V_1 + V_2)^2 \quad (27.24)$$

Similarly, the squaring current in the difference-square circuit of Fig. 27.28b is given by

$$I_{SQ(b)} = \frac{\beta_{23}}{4}(V_1 - V_2)^2 \quad (27.25)$$

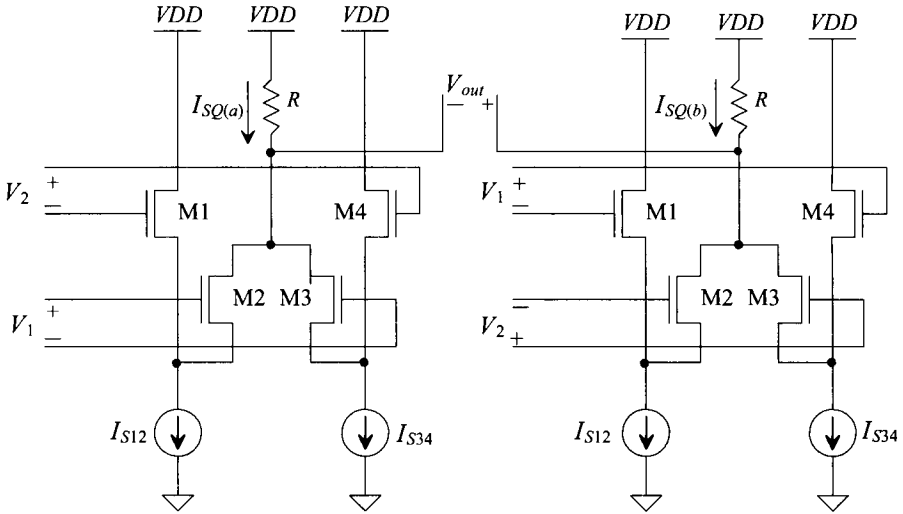


Figure 27.28 (a) Sum-squaring circuit and (b) difference squaring circuit.

The output voltage of the sum-square circuit is given by

$$V_{o-} = VDD - I_{SQ(a)}R \quad (27.26)$$

while the output voltage of the difference-square circuit is given by

$$V_{o+} = VDD - I_{SQ(b)}R \quad (27.27)$$

A multiplier is formed by taking the difference between these voltages. The output voltage of the multiplier of Fig. 27.27 is given by

$$V_{out} = V_{o+} - V_{o-} = R \frac{\beta_{23}}{4} [(V_1 + V_2)^2 - (V_1 - V_2)^2] \quad (27.28)$$

or, using Eq. (27.23)

$$V_{out} = R\beta_{23} \cdot V_1 V_2 \quad (27.29)$$

The fundamental concern with using this type of multiplier is the fact that modern short-channel CMOS devices don't follow the square-law equations (Eq. [27.24]) we used to derive this result.

ADDITIONAL READING

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PROBLEMS

- 27.1** Using the long-channel CMOS process, compare the performance (using simulations) of the comparator in Fig. 27.8 with the comparator in Fig. 27.9. Your comparison should include DC gain, systematic offset, delay, sensitivity, and power consumption.
- 27.2** Show, using simulations, how the addition of a balancing resistor in Fig. 27.14 can be used to improve the response seen in Fig. 27.13.
- 27.3** Simulate the operation of the comparator in Fig. 27.15 in the short-channel CMOS process. Determine the comparators sensitivity and the kickback noise.
- 27.4** Repeat problem 27.3 for the comparator in Fig. 27.16. Show that the input common-mode range of the comparator in Fig. 27.16 extends beyond the power supply rails.
- 27.5** Simulate the operation of the input buffer in Fig. 27.17 in the short-channel CMOS process. How sensitive is the buffer to input slew-rate? How symmetrical are the output rise and fall times? Suggest, and verify with simulations, a method to reduce the power consumed by the input buffer.
- 27.6** Design a low power clocked comparator for use with a Flash ADC (discussed in Ch. 29). Use the short-channel CMOS process and a clocking frequency of 250 MHz estimate the power dissipated by 256 of these comparators.

Data Converter Fundamentals

Data converters (a circuit that changes analog signals to digital representations or vice-versa) play an important role in an ever-increasing digital world. As more products perform calculations in the digital or discrete time domain, more sophisticated data converters must translate the digital data to and from our inherently analog world. This chapter introduces concepts of data conversion and sampling which surround this useful circuit.

28.1 Analog Versus Discrete Time Signals

Analog-to-digital converters, also known as A/Ds or ADCs, convert analog signals to discrete time or digital signals. Digital-to-analog converters (D/As or DACs) perform the reverse operation. Figure 28.1 illustrates these two operations. To understand the functionality of these data converters, it would be wise first to compare the characteristics of analog versus digital signals.

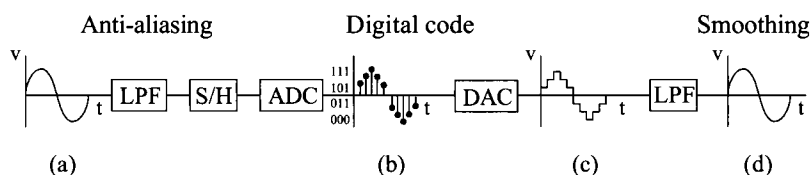


Figure 28.1 Signal characteristics caused by A/D and D/A conversion.

In Fig. 28.1 the original analog signal (a) is filtered by an anti-aliasing filter to remove any high-frequency components that may cause an effect known as aliasing (see Sec. 28.5). The signal is sampled and held and then converted into a digital signal (b). Next the DAC converts the digital signal back into an analog signal (c). Note that the output of the DAC is not as “smooth” as the original signal. A low-pass filter returns the analog signal back to its original form (plus phase shift introduced from the conversions)

after eliminating the higher order signal components caused by the conversion. This example illustrates the main differences between analog and digital signals. Whereas the analog signal in Fig. 28.1a is *continuous* and *infinite* valued, the digital signal in (b) is *discrete* with respect to time and *quantized*. The term *continuous-time signal* refers to a signal whose response with respect to time is uninterrupted. Simply stated, the signal has a continuous value for the entire segment of time for which the signal exists. By referring to the analog signal as infinite valued, we mean that the signal can possess any value between the parameters of the system. For example, in Fig. 28.1a, if the peak amplitude of the sine wave was +1V, then the analog signal can be any value between -1 and 1 V (such as 0.4758393848 V). Of course, measuring all the values between -1 and 1 V would require a piece of laboratory equipment with infinite precision.

The digital signal, on the other hand, is discrete with respect to time. This means that the signal is defined for only certain or discrete periods of time. A signal that is quantized can only have certain values (as opposed to an infinitely valued analog signal) for each discrete period. The signal illustrated in Fig. 28.1b illustrates these qualities.

28.2 Converting Analog Signals to Digital Signals

We have already established the differences between analog and digital signals. How is it possible to convert from an analog signal to a digital signal? An example will illustrate the process.

Where you live the temperature in the winter stays between 0° F and 50° F (Fig. 28.2a). Suppose you had a thermometer with only two readings, hot and cold, and you wanted to record the weather patterns and plot the results. The two quantization levels can be correlated with the actual temperature as follows:

If $0^{\circ} \text{ F} \leq T < 25^{\circ} \text{ F}$	Temperature is recorded as cold
If $25^{\circ} \text{ F} \leq T < 50^{\circ} \text{ F}$	Temperature is recorded as hot

You take a measurement every day at noon and plot the results after one week. From Fig. 28.2b, it is apparent that your discretized version of the weather is not an accurate representation of the actual weather.

Now suppose that you find another thermometer with four possible temperatures (hot, warm, cool, and cold) and you increase the number of readings to two per day. The result of this reading is seen in Fig. 28.3a. The quantization levels represent four equal bands of temperature as seen below:

If $0^{\circ} \text{ F} \leq T < 12.5^{\circ} \text{ F}$	Temperature is recorded as cold
If $12.5^{\circ} \text{ F} \leq T < 25^{\circ} \text{ F}$	Temperature is recorded as cool
If $25^{\circ} \text{ F} \leq T < 37.5^{\circ} \text{ F}$	Temperature is recorded as warm
If $37.5^{\circ} \text{ F} \leq T < 50^{\circ} \text{ F}$	Temperature is recorded as hot

Here, the digital version of the weather still looks nothing like the actual weather pattern, but the critical issues in digitizing an analog signal should be apparent. The actual weather pattern is the analog signal. It is continuous with respect to time, and its value can be between 0° F and 50° F (even 33.9638483920398439° F!). The accuracy of the

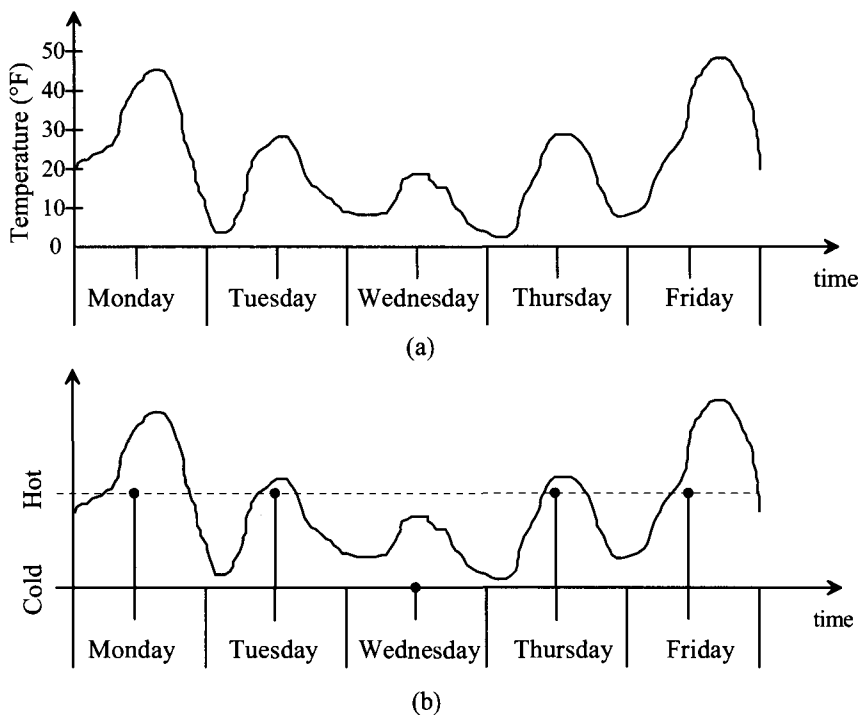


Figure 28.2 (a) An analog signal representing the temperature where you live and (b) a digital representation of the analog signal taking one sample per day with two quantization levels.

digitized signal is dependent on two things: the number of samples taken and the resolution, or number of quantization levels, of the converter. In our example, we need to increase both the number of samples and the resolution of thermometer.

Suppose that finally we obtain a thermometer with 25 temperature readings and that we take a reading eight times per day. Each of the 25 quantization levels now represents a 2° F band of temperature. From Fig. 28.3b, we can see that the digital version of the weather is approaching that of the actual analog signal. If we kept increasing both sampling time and resolution, the difference between the analog and the digital signals would become negligible. This brings up another critical issue: how many samples should one take in order to accurately represent the analog signal?

Suppose a sudden rainstorm swept through your town and caused a sharp decrease in temperature before returning to normal. If that storm had occurred between our sampling times, our experiment would not have shown the effects of the storm. Our sampling time was too slow to catch the change in the weather. If we had increased the number of samples, we would have recognized that something happened which caused the temperature to drop dramatically during that period.